Initial Post - Applications of AI: A Financial Services Industry Perspective

From virtual assistants, personal recommendation systems and smart home devices to fraud detection systems, healthcare and even educational support systems, the applications of Artificial Intelligence (AI) have become the *status quo* in daily life. This is due to increased computing power to support AI models and the availability of large datasets. Since the introduction of the world wide web and in more recent times—that is, the last 10-15 years—the quantity of data available online is in abundance which supports Kneusel's (2023) assertion that "awe-inspiring things can happen."

The financial services (FS) industry is the prime example of such a data enriched industry that is readily adopting AI practices to aid in operational efficiencies, cost reductions, and supporting banks with compliance and regulatory responsibilities (Douglas, Barberis and Buckley, 2017). As a result, core banking systems and regulatory software providers are integrating AI into their software solutions. For example, Temenos (2024) has integrated Generative AI (GenAI) capabilities, termed Explainable AI (XAI) solutions, into their core banking and Financial Crime Mitigation (FCM) systems. The key ethos of Temenos' XAI solution is explainability; thus, aligning to the core principles of ethical AI practice—transparency and repeatability.

The drawback of AI use in the FS industry suggests workplace displacement may become normalised. Theoretically, humans are replaceable by AI; however, the recent Deloitte scandal in 2025 where GenAI use resulted in AI hallucinations being present in a report published by the Australian government (Paoli, 2025) proves that humans cannot be fully replaced by AI. This emphasises the importance for human in the loop mechanisms to monitor and validate AI generated output. Arguably, GenAI could become somewhat self-regulated by introducing symbolic systems to detect hallucinations; however, the implementation of such an AI system is currently unknown (Kneusel, 2023).

Additionally, a human-centred AI approach is necessary. This is supported by the ethical use of AI (and data) which needs to be aligned with local and international regulations such as GDPR, ensuring bias in AI models is mitigated and customer data is not misused.

In conclusion, the strategic implementation of AI is a necessary tool for operational efficiencies and reporting processes in the FS industry. However, banks need to address ethical issues, workforce displacement and issues associated with hallucinations while implementing and using AI.

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Peer Response 1

Hi Aldana - thank you for reviewing the automative industry (Tesla). When I think of AI, I often forget that this industry is probably one of the frontrunners of AI innovation.

Your post provides a critical overview of the ways in which AI driven innovations are enhancing operational efficiencies and consumer trust. Using Tesla as an example of an organisation embracing and implementing AI solutions in the forms of accident detection, self-driving vehicles and speech command interfaces further supports your argument.

The post clearly details the ethical implications of AI use at Tesla. However, your analysis could be straightened further by including an analysis of the legal implications of AI use, with respect to the Tesla EV accident (Etzioni and Etzioni, 2017) and determining who is responsible for the accident — the driver or Tesla? The Automated Vehicles Act 2024 details the potential forms of liability in an automated vehicle accident; these include the manufacturers liability, third party liability, service providers liability and, if course, the driver's liability (Lyons Davidson, 2024).

Further consideration of the ways in which Tesla's approach aligns with SDGs could be incorporated to enhance your work further by consider the balancing act of considering environmental benefits vs being one of the frontrunners in AI-drive technology innovations.

I agree with your conclusion—the ethical concerns raised by AI-use need to be adequately regulated from an ethical perspective. Additionally, safeguards need to be in place to ensure further accidents such as the above mentioned EV accident are mitigated.

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Peer Response 2

Hi Piotr, thank you for your post which has informed my naive self of the use of AI in the legal field.

Your post examines the usefulness of NLP being used to automate labour intensive roles such as e-discovery and due diligence.

Although, I can see this being extremely beneficial financially, I do question the integrity of the work being performed as we are well versed in how common AI hallucinations can be, for example, Deloitte Australia using GenAI for a report that contained hallucinations (Paoli, 2025). Perhaps considering the implications of AI hallucinations would enhance this piece of work & factoring in the need for human in the loop review processes to validate the AI output?

You also make reference to the use of AI to draft legal documents and contracts. This is indeed quite compelling, as you rightly assert that this levels the professional playing filed for junior lawyers versus their more experienced senior counterparts.

Your conclusion is quite clear in that the role of the lawyer is being re-evaluated, which I 100% agreed with. A point to note is that AI is empowering non-legally trained individuals to represent themselves in more straightforward legal cases as highlighted by Perlo and Yan (2025) in their recent NBC News reports. I wonder if acknowledging this recent trend will strengthen your argument as, this, coupled with the more level playing field between junior and senior lawyers, is essentially de-skilling the role of a lawyer (I am probably oversimplifying here)?

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Peer Response 3

Hi Yasmin - thank you for your post which addresses AI use in an industry that we all probably take for granted with the concept of “googling” becoming the normal in our daily lives.

Your post presents a strong case for AI being the driving force for increased business value; this concept is illustrated clearly using the example of Google as your case study.

The post details how AI has benefited Google from an economic perspective, and also questions if businesses in other industries should adopt AI to keep up with technological advancement. Thus supporting the concept that “awe-inspiring things can happen” (Kneusel, 2023). We even see Google competing with OpenAI’s ChatGPT with Google Gemini (Google AI) being added as an option on the Google search engine.

In my opinion, the reason for Google’s success is due the abundance of data available on the online. However, I think further consideration of the ethical implications of using this data will strengthen the post such as data governance regulations at a local and global level (e.g. GDPR). This, of course, is an awkward concept to address at Google since I am sure they are navigating the tricky landscape of conflicting regulations due to the global nature of Google’s business (i.e. fully online, with data crossing global borders).

I agree with your conclusion as companies like Google need to embrace AI in order to stay competitive in such a quickly moving technological landscape.

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Summary Post - AI in Financial Services (FS)

My initial post examined the embedding of AI into the FS industry, highlighting benefits such as cost reductions and process improvements, while also addressing the ethical challenges associated with AI, workplace displacement and AI hallucinations.

Peer feedback reinforced the need for transparency for AI usage, particularly regarding data usage in credit-scoring and ensuring decision-making algorithms are free from bias. Consequently, explainable AI should positioned at the forefront of any AI solution design and implementation. This is emphasised by Peres *et al*. (2020, p. 220134) where they state that explainability is key to influencing stakeholder decision-making during AI implementation. This further strengthens my initial argument for the need of human-in-the-loop mechanisms to validate and interpret AI-generated outputs.

Reviewers have questioned if re-skilling alone can address workplace displacement; I suspect that new job roles that require human oversight and domain expertise (i.e. human-in-the-loop mechanisms) will replace the more low value roles. Thus reinforcing the view that workplace displacement is a genuine concern, mirroring patterns during the Industrial Revolution (Race, 2025). Furthermore, evidence suggests a correlation between unemployment and AI implementation (Ozkan and Sullivan, 2025), a pattern that will become normalised as companies prioritise automation. Recent lay-offs by major companies further indicate that re-skilling alone is not a viable solution for workplace displacement (Lichtenberg, 2025; Kaye, 2025).

One reviewer noted that while the prospect of using symbolic AI for detecting hallucinations is an interesting concept, it would be difficult to implement. This supports my own views as a self-regulating AI system would essentially be a form of artificial general intelligence (AGI), which, is implausible due to current technological limitations. Moreover,

this conflicts with the principles of Industry 5.0 which emphasises human and AI collaboration rather than full AI automation (Kraaijenbrink, 2022).

In conclusion, although different industries have differing regulatory approaches, Leslie's (2019) key principles—these are, Fairness, Accountability, Sustainability, and Transparency—remain universally applicable to responsible AI governance.

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